

FinTrack - Personal Finance DashBoard and Predictive Analytics



Minor Project Report Submission by :

Sandeep Singh (23124094)
Manan Joshi (23124057)
Vanshika (23124118)
Mayank Sharma (23124061)

Under the Mentorship of

Dr. Avani Vyas

Assistant Professor,

Department of Information Technology

**Dr B R Ambedkar National Institute of Technology
Jalandhar-144008, Punjab (INDIA)**

Declaration

We, the undersigned, hereby solemnly declare that the Project Report titled “**FinTrack - Personal Finance Dashboard and Predictive Analytics**” submitted in partial fulfillment of the requirements for the award of the **Bachelor of Technology (B.Tech) in Information Technology**, is an authentic and original work carried out by us during the academic year 2025-2026 under the guidance and supervision of **Dr. Avani Vyas , Assistant Professor, Department of Information Technology**, Dr B R Ambedkar National Institute of Technology Jalandhar.

This project work is a result of our own efforts, study, research, and observations. It has not been submitted, either in part or full, to any other University, Institute, or Organization for the award of any degree, diploma, certificate, or published in any form of academic or research journal.

We further affirm that all the information, facts, data, diagrams, and results presented in this report are genuine and have been gathered from reliable sources. Any material reproduced or referred to from external sources such as books, research papers, websites, or articles has been clearly acknowledged and cited in the References section of this report as per academic integrity norms.

We understand that plagiarism or false claims in academic work is a serious offense and accept full responsibility for the authenticity and originality of the contents of this project report.

Team Members:

Sandeep Singh (Roll No. 23124094)

Manan Joshi (Roll No. 23124057)

Vanshika (Roll No. 23124118)

Mayank Sharma (Roll No. 23124061)

Certificate

This is to certify that the project report entitled **“Personal Finance Dashboard and Predictive Analytics”** submitted by Sandeep Singh (Roll No. 23124094) , Manan Joshi (Roll No. 23124057), Vanshika (Roll No. 23124118) Mayank Sharma (Roll No. 23124061) to Dr B R Ambedkar National Institute of Technology Jalandhar, in partial fulfillment for the award of the degree of B. Tech in Information Technology has been carried out under our supervision and that this work has not been submitted elsewhere for a degree.

Dr. Avani Vyas
Assistant Professor
Department of Information Technology

Acknowledgment

With great pleasure and sincerity, we take this opportunity to express our profound gratitude and deep regards to all those who supported, guided, and encouraged us throughout the completion of our project titled “**Personal Finance Dashboard and Predictive Analytics.**”

This project has been a significant learning experience both academically and personally, and without the valuable contributions of several individuals, its completion would not have been possible.

First and foremost, we would like to extend our deepest gratitude to our respected guide, **Dr. Avani Vyas , Assistant Professor, Department of Information Technology**, whose continuous guidance, expert supervision, insightful suggestions, and constructive feedback have been instrumental in shaping this project. Their encouragement and motivation kept us focused and committed during every phase of development.

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Team Members:

Sandeep Singh (Roll No. 23124094)

Manan Joshi (Roll No. 23124057)

Vanshika (Roll No. 23124118)

Mayank Sharma (Roll No. 23124061)

Abstract

The FinTrack application is a modern personal finance management system designed to assist users in effectively tracking, analyzing, and optimizing their financial activities. With the increasing complexity of daily expenses and financial commitments, individuals often find it difficult to maintain a structured record of their income, spending, and savings. This project aims to solve that problem by providing a centralized and intelligent platform for financial management.

The system is developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js), which ensures high performance, scalability, and a responsive user interface. Users can easily add, update, and delete transactions, categorize them, and view a real-time summary of their financial status through an interactive dashboard.

A key highlight of the application is the integration of advanced financial analytics. The system implements the 50-30-20 budgeting rule, enabling users to compare their actual spending patterns with ideal financial distribution. Additionally, it includes a budget tracking feature with visual indicators to alert users when they exceed their limits.

The project also introduces intelligent features such as automatic monthly rollover detection and a financial goal tracker. The goal tracker uses historical transaction data to estimate the time required to achieve specific financial targets, thereby encouraging disciplined saving behavior.

Overall, FinTrack not only records financial data but also transforms it into meaningful insights, empowering users to make informed financial decisions, improve spending habits, and work towards long-term financial stability.

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Chapter 1 : Introduction

1.1 Background of the Study

Managing finances is one of the most crucial parts of people's lives. Most people find it difficult to keep track of all their expenses, savings, budgets, and overall financial state. Using conventional techniques such as notebooks and spreadsheet files is very time-consuming and does not involve any form of automation that can be very useful when managing finances. In light of the increasing trend towards digitalization, creating a Personal Finance Dashboard that uses the latest technologies, such as MERN stack (MongoDB, Express.js, React.js, and Node.js), becomes highly important.

A Personal Finance Dashboard will help people record and categorize all their incomes and expenditures (food, travel, bills, shopping, education, and so on).

1.2 Problem Statement

People face the following challenges:

- Difficulty in managing expenses and income manually
- No real-time tracking of budget and spending habits
- No categorized insights into where money is being spent
- No digital tool for monitoring financial status interactively

1.3 Objectives

1.3.1 General Objective

To develop a web-based Personal Finance Dashboard using MERN Stack to help users record, manage, and analyze their financial data effectively.

1.3.2 Specific Objectives

- Build a digital application to track income and expenses.
- Categorize expenses into meaningful sections.

- Provide visual representation using charts and graphs.
- Offer a secure login and authentication system using JWT.
- Make the system responsive and accessible from any device.

1.4 Scope of the Project

Included:

- To develop a web-based personal finance dashboard using MERN stack
- To track and categorize income and expenses efficiently
- To implement intelligent financial analysis using the 50-30-20 rule
- To develop a goal tracking and savings prediction system
- To introduce a monthly rollover mechanism for financial continuity
- To provide real-time budget monitoring with visual indicators

1.5 Significance of the Study

Area	Purpose
Students	Learn MERN stack, Data visualization, Web app development
General Users	Manage, analyse and predict monthly financial activities digitally
Developers	Learn secure full-stack application with JWT, MongoDB
Institutions	Can enhance it to an expense tracker or fintech product

Chapter 2 : Existing System

2.1 Financial Tracking Systems

Earlier, expense tracking was done manually through spreadsheets and notebooks and was not automated, categorised or visualised through graphs. Today, finance management systems on the web are more user-friendly and improved visualisation .

2.2 Dashboard-Based Finance Systems

A finance dashboard will show an interface which consists of data such as the expenses incurred, income earned, various categories, trends and report. These kinds of interfaces assist people to understand better the spending trends using graphical representation.

2.3 Importance of MERN in Financial Web Applications

Technology	Purpose
MongoDB	Stores financial records (income, expense, category, date)
Express.js	Backend routing, API management
React.js	Frontend UI, dashboard components
Node.js	Server-side JavaScript runtime

2.4 Limitations of Existing Systems

Despite advancements, current financial systems still face several limitations:

- Lack of intelligent budgeting techniques
- No implementation of standard financial rules (like 50-30-20)
- Absence of goal prediction and financial planning features
- Limited automation in handling recurring or monthly financial changes
- No integration of historical data for future insights

2.5 Need for Improved System

Due to the above limitations, there is a need for a more advanced financial management system that:

- Provides intelligent analysis along with data tracking
- Implements structured financial rules
- Includes predictive features for goal achievement
- Automates processes like monthly rollover
- Enhances user experience through interactive dashboards

Chapter 3 : System Analysis

3.1 System Requirements

3.1.1 Hardware Requirements

- Processor – i3 and above
- Minimum 4GB RAM
- 500GB HDD
- Internet Connectivity

3.1.2 Software Requirements

- Python, Flask/Django
- MySQL- PostgreSQL /MongoDB
- Power BI-NLP / MS Excel / Plotly
- Windows/Linux/Mac OS

3.2 Feasibility Study

- **Technical Feasibility** – All required technology is freely available.
- **Operational Feasibility** – System easy for non-technical users.
- **Economic Feasibility** – Low-cost or free execution using open-source tools.

Chapter 4 : System Design

4.1 System Design Overview

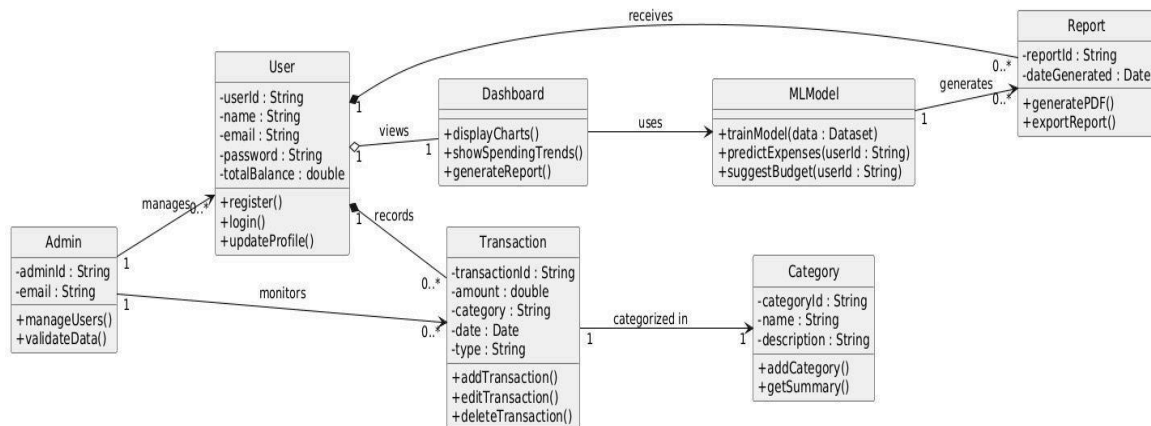
The system is built using the **client-server model** where React.js acts as the client side, Node.js/Express.js serves as the backend, and MongoDB stores all financial data.

4.2 System Architecture

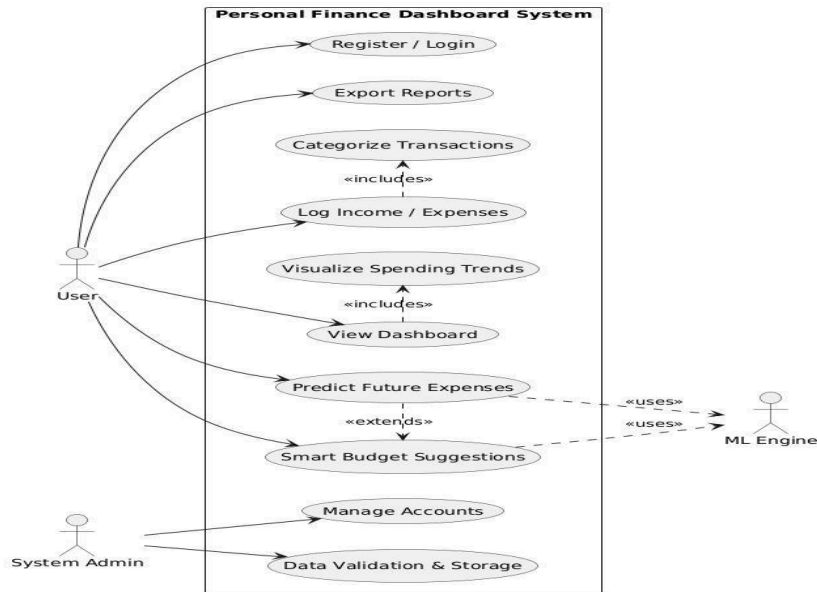
Layer	Technology	Purpose
Presentation	React.js	Forms, Charts, UI
Business Logic	Node.js & Express	API, routing, validation
Data Layer	MongoDB	Stores income/expenses
Security	JWT, Bcrypt	Authentication & authorization

4.3 UML Diagram

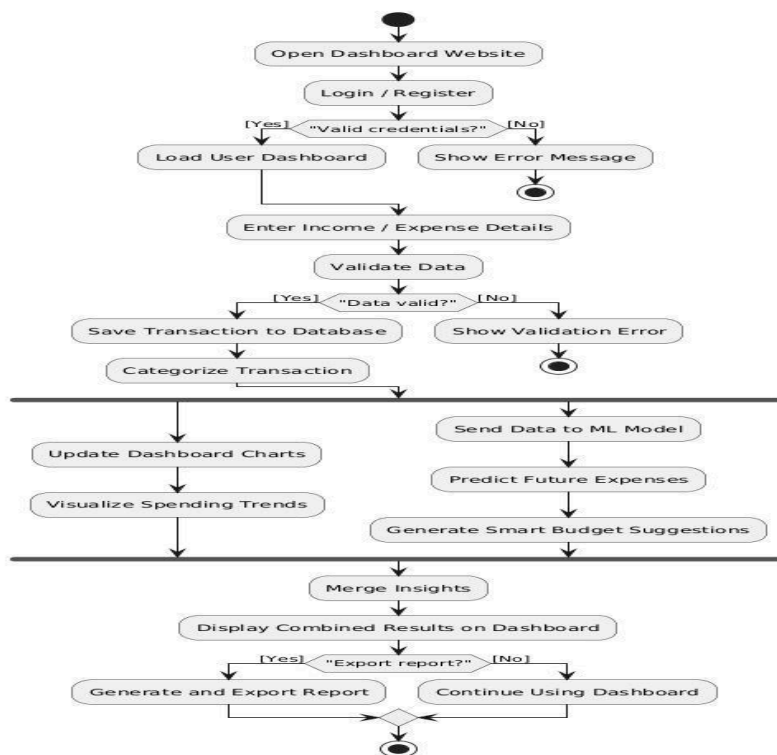
Class Diagram:



Use Cases Diagram:



Activity Diagram:



Chapter 5 : SYSTEM IMPLEMENTATION

5.1 System Overview

The system is a web-based personal finance dashboard where users can record, manage, visualise and track income, expenses, budgets and account balances. It supports authentication, dashboards, categorisation, budget alerts, and basic reports — but doesn't yet offer forecasting or ML-based predictive analytics.

5.2 MERN Technology Utilization

Component	Technology Used	Purpose
Frontend	React JS	User Interface & Data Visualization
Backend	Node.js + Express	API, Business Logic, Request Handling
Database	MongoDB (Atlas)	Stores Users, Transactions, Accounts, Budgets
Authentication	JWT + bcrypt	Secure Login, Token-based Auth

5.3 Core Modules

1. User Authentication Module

- User signup, login, logout
- Password hashing, JWT token-based authentication
- Manage user profile settings (currency, time-format, profile info)

2. Account Management Module

- Add, update, delete financial accounts (Bank, Wallet, Cash)
- Track balance per account
- Set default account for transactions

3. Category and Budget Module

- Create expense/income categories (Groceries, Rent, Salary, etc.)
- Set and monitor monthly budgets per category
- Notify when budget exceeds limit

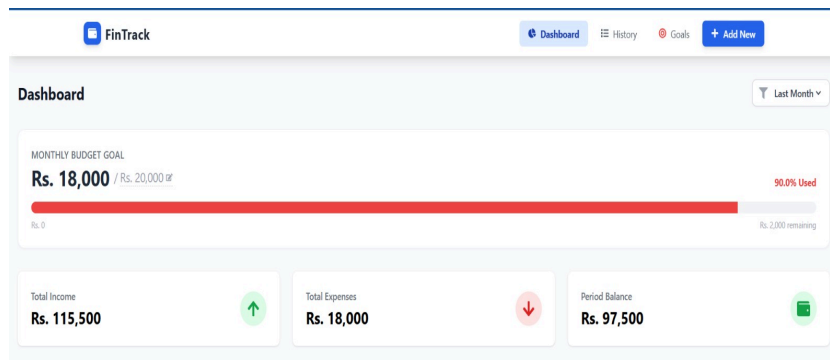
4. Transaction Management Module

- Record income and expenses manually
- Categorization and tagging of each transaction
- Filter by date, type, category, amount
- Manage recurring transactions

The screenshot shows the 'Add New Transaction' form in the FinTrack application. The form is titled 'Add New Transaction' and has a 'Back' button. It contains a 'TRANSACTION IMPACT' section showing 'Current Balance' as Rs. 97500.00 and 'After Transaction' as Rs. 97500.00. Below this, there are input fields for 'Title', 'Amount' (set to Rs. 0.00), 'Category' (set to Food), 'Type' (set to Expense), and 'Date' (set to 03/24/2026). At the bottom, there are 'Reset' and 'Add Transaction' buttons.

4. Dashboard and Visualization Module

- Overview of financial status
- Visual charts (Bar, Pie, Line) showing spending patterns
- Monthly income vs. expenses comparison
- Category-wise expense breakdown



5. Reports and Export Module

- View monthly summary
- Filtered reports by date/category
- Export transactions as CSV or PDF

5.4 Advanced Financial Analytics Module

1. 50-30-30 Rule

The system is improved with smart financial analysis tools that go beyond simple transaction tracking. It allocates expenses into Needs (50%), Wants (30%) and Savings (20%) based on predefined category mapping.

Code Implementation:

```
const CATEGORY_MAPPING = {  
  
  Food: 'Needs',  
  
  Housing: 'Needs',  
  
  Utilities: 'Needs',  
  
  Healthcare: 'Needs',  
  
  Transport: 'Needs',  
  
  Education: 'Needs',  
  
  Shopping: 'Wants',  
  
  Entertainment: 'Wants',  
  
  Savings: 'Savings',  
  
  Investment: 'Savings'  
  
};
```



2. Monthly Rollover System

The system automatically detects a change in month and calculates the remaining balance from the previous month.

Code Implementation:

```
const lastSeenMonthKey =  
localStorage.getItem('lastSeenMonth');  
  
const currentMonthKey =  
`${now.getFullYear()}-${now.getMonth()}`;  
  
if (lastSeenMonthKey && lastSeenMonthKey !==  
currentMonthKey) {  
  
}
```

3. Goal Tracking and Prediction

Users can set financial goals and track their progress.

- **Monthly Savings Calculation:**

```
const avgSavings = totalSavings / validMonths;
```

- **Prediction Formula:**

```
const monthsNeeded = Math.ceil((target - saved) /  
monthlySavings);
```

Chapter 6 : Conclusion

The FinTrack project successfully demonstrates the development of a comprehensive and intelligent personal finance management system that goes beyond basic expense tracking. The system combines modern web technologies and practical financial analysis tools, giving users control and insight into their financial activities.

Throughout the project key functionalities such as transaction management, budget tracking and category based analysis have been implemented effectively. The 50-30-20 rule adds an element of standard budgeting to the system, allowing users to compare their financial behaviour to general budgeting practices and making the system informative as well as educational.

One of the major contributions of this work is the incorporation of smart features such as monthly rollover handling, goal prediction. The rollover feature allows you to continue your financial tracking from month to month, while the goal tracker enhances user motivation by providing realistic timelines based on actual savings patterns. These features make the system more dynamic and user-centric.

On the technical side, the MERN stack is used to ensure a modular, scalable and efficient architecture. The frontend provides a responsive and intuitive user experience, while the backend ensures secure data storage and smooth API communication.

To sum up, FinTrack is a powerful financial assistant, which not only helps users to track their finances, but also guides them towards making better financial decisions. The project has a lot of potential for future upgrades such as AI based financial recommendations, real time banking integration, and mobile application development. The project is a scalable and impact solution in the domain of personal finance management.

Chapter 7 :Future Scope

1. Predictive Analytics (AI/ML Integration)

- Utilizing Machine Learning algorithms like Linear Regression, ARIMA, and LSTM for:
 - Expense forecasting
 - Budget recommendation
 - Saving predictions
 - Debt management suggestions

2. Integration with Bank APIs

- Automatic tracking of transactions directly from users' bank accounts using:
 - Plaid API
 - PayPal API
- Real-time balance updates and automatic category classification.

3. Mobile App Version (React Native)

- Extend the dashboard into a cross-platform mobile app.
- Push notifications for bill reminders, budget alerts, savings milestones.

4. Enhanced Security Features

- Implement Multi-Factor Authentication (MFA)
- Biometric login (fingerprint/face)
- Encrypted financial data storage using AES/DES

5. Collaborative Family Finance Management

- Multiple user roles (Parent, Child, Partner)
- Shared financial goals and budget planning

8. Automatic Expense Categorization

- Use NLP to categorize expense text (Fuel, Rent, Food, Bills)
- Smart tagging using AI

Chapter 8: References

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